Siemens ACM200 Axle Counter Sensor Head Alignment

Alignment Tolerances, Mounting Torques, and Vibration Monitoring

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1. Sensor Head Overview

The ACM200 wheel sensor head detects passing wheel flanges using an inductive electromagnetic field. Correct alignment relative to the rail running surface is critical for reliable axle counting. Misalignment can result in missed counts (wheel passes undetected) or double-counts (single axle counted twice), both of which create safety-critical track occupancy errors. The sensor is mounted on an adjustable bracket bolted to the rail web.

[Schematic: ACM200 sensor head mounting on rail web - alignment axes]

Figure: ACM200 sensor head mounting on rail web - alignment axes

2. Alignment Specifications

Alignment Parameter	Required Value	Measurement Method
Lateral position (centre of rail)	+/- 2 mm from running surface centreline	Alignment gauge
Vertical height above rail foot	25 mm to 32 mm	Height gauge
Longitudinal angle to rail	90 degrees +/- 2 deg	Digital angle gauge
Cable strain relief clearance	> 50 mm free cable	Visual inspection

WARNING: Misalignment beyond the tolerances above must be corrected immediately. An out-of-tolerance sensor should be treated as a provisional fault until re-aligned and a full axle count verification cycle has been completed.

3. Alignment Procedure

- Step 1: Obtain track protection and ensure the line is blocked.
- Step 2: Loosen the four M10 mounting bolts on the sensor bracket (do not fully remove). Slacken only enough to allow lateral and vertical adjustment.
- Step 3: Using the alignment gauge, position the sensor at the correct lateral offset (within +/- 2 mm of the rail centreline). Tighten the lateral locking bolt hand-tight.
- Step 4: Using the height gauge, set the sensor at 28 mm above the rail foot (mid-point of the 25 - 32 mm tolerance). This centre-point setting provides the best margin against track movement and sensor drift.
- Step 5: Verify angular alignment with the digital angle gauge. Adjust if needed.
- Step 6: Torque all four M10 mounting bolts to 25 Nm in a cross pattern. Recheck all three alignment dimensions after torquing.
- Step 7: Perform a full axle count verification: have a track vehicle (or manually simulate with a test axle) pass through the section. Confirm the count matches expected.

NOTE: Mounting bolt torque of 25 Nm is critical. Under-torqued brackets can loosen under traffic vibration, leading to progressive misalignment. Check torque at every scheduled maintenance visit (every 6 months minimum).

4. Vibration Monitoring

The ACM200 sensor includes an internal MEMS vibration sensor. Sustained vibration readings above 4.5 g indicate that the mounting bracket has loosened or that an abnormal vibration source is present (e.g. flat-spotted wheel on a frequent train path). The following action thresholds apply:

Vibration Level	Duration	Action
< 3.5 g	Any	Normal - no action required
3.5 - 4.5 g	Sustained (> 5 min)	Check bracket bolt torques at next visit
> 4.5 g	Any sustained reading	Inspect and torque-check M10 bolts to 25 Nm within 48
> 6.0 g	Any reading	Inspect immediately; may indicate structural bracket fai